News Recommendation System

COMP 9727

Media Mogul

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Contents

Introduction

- News recommendation has been widely studied.
- Baseline, feature-based methods (LibFM, DSSM …) lack the performance of more complex methods incorporating user data (GRU, DKN …).[1]
- However, user-based methods such as CF or temporal based either perform poorly or lack a study into their models' cold-start performance.[2][3]
- This model performance gap motivates our development of a new news recommender model.

	Overall			
	AUC	MRR	nDCG@5	nDCG@10
LibFM	59.93	28.23	30.05	35.74
DSSM	64.31	30.47	33.86	38.61
Wide&Deep	62.16	29.31	31.38	37.12
DeepFM	60.30	28.19	30.02	35.71
DFM	62.28	29.42	31.52	37.22
GRU	65.42	31.24	33.76	39.47
DKN	64.60	31.32	33.84	39.48
NPA	66.69	32.24	34.98	40.68
NAML	66.86	32.49	35.24	40.91
LSTUR	67.73	32.77	35.59	41.34
NRMS	67.76	33.05	35.94	41.63

Figure: Comparison of competitor performance.

[1] Wu, F., Qiao, Y., Chen, J.-H., Wu, C., Qi, T., Lian, J., Liu, D., Xie, X., Gao, J., Wu, W., & Zhou, M. (2020). MIND: A large-scale dataset for news recommendation. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics* (pp. 3597–3606). Association for Computational Linguistics. https://doi.org/10.18653/v1/2020.acl-main.331 ACL Anthology
[2] Wang, H., Zhang, F., Xie, X., & Guo, M. (2018). DKN: Deep knowledge-aware network for news recommendation. In *Proceedings of the 2018 World Wide Web Conference* (pp. 1835–1844). ACM.
[3] Wu, C., Wu, F., Ge, S., Qi, T., Huang, Y., & Xie, X. (2019). Neural news recommendation with multi-head self-attention. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing and the 9th International Joint Conference on Natural Language Processing* (EMNLP-IJCNLP) (pp. 6389–6394). Association for Computational Linguistics. Retrieved from https://aclanthology.org/D19-1671

Introduction

Research Questions

- How can we best build a news recommender that possesses superior performance by leveraging user data while retaining the cold-start performance of feature-based systems.
  - Performance defined as a prediction and ranking problem.
  - **Predict** what articles a user may be interested in and **rank** them, to determine what comes on the front page.

Cold-start Issues

- New community/database.
  - Low amount of information available for methods relying on user data.
- New User.
  - No information regarding user, therefore not possible to provide recommendations yet.
- Assumptions**:**
  - Assume that news articles are always available.

System Architecture

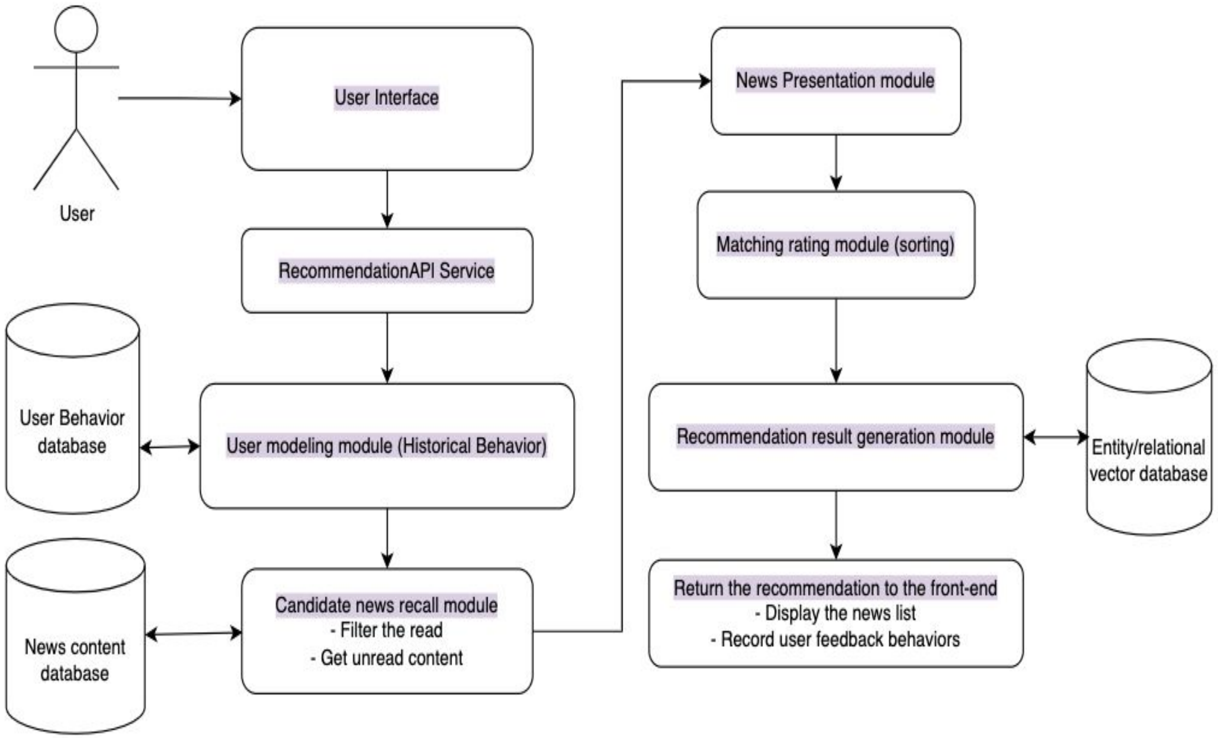


Figure: The news recommender would be deployed behind a user interface.

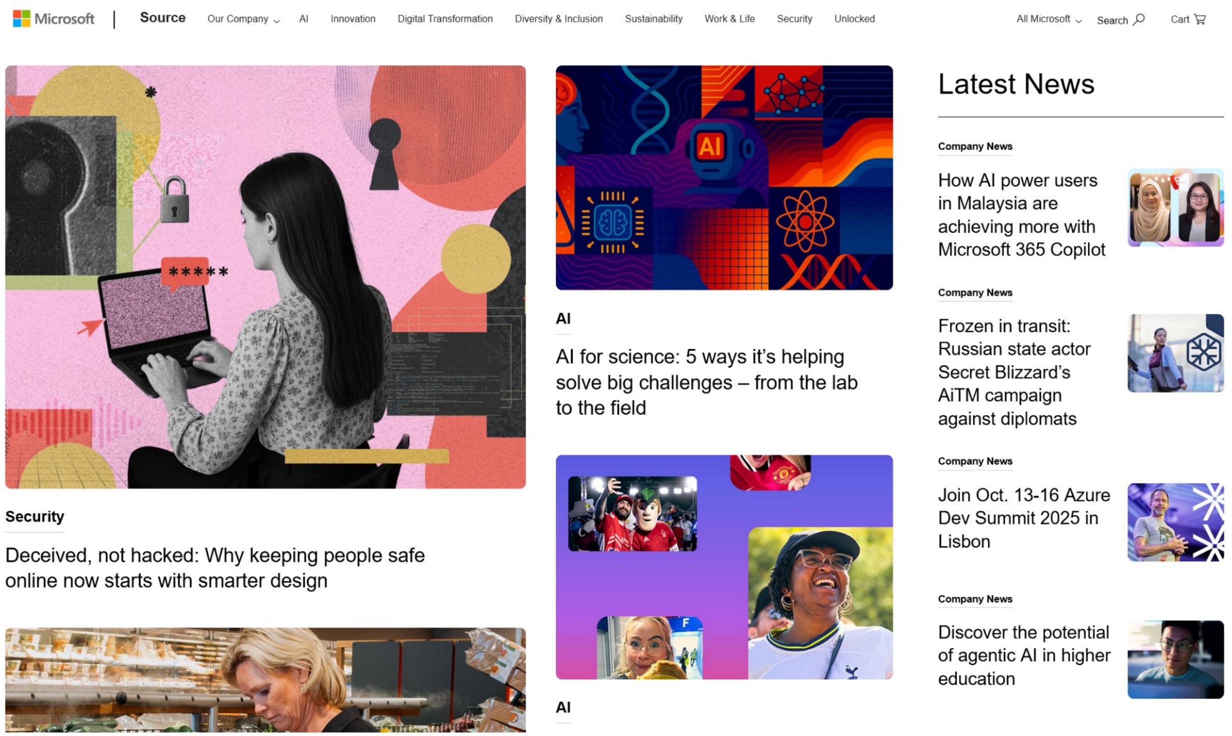


Figure: Mockup of the UI system

Dataset

Dataset - MIND news

The MIND dataset for news recommendation was collected from anonymised behavior logs of Microsoft News website. The dataset randomly sampled 1 million users who had at least 5 news clicks during 6 weeks from October 12 to November 22, 2019.

- Behaviors.tsv - The behaviors.tsv file contains the impression logs and users' news click histories.156965 rows

```
behaviors.head()
```

	Impression_ID	User_ID	Time	History	Impressions
0	1	U13740	11/11/2019 9:05:58 AM	N55189 N42782 N34694 N45794 N18445 N63302 N104...	N55689-1 N35729-0
1	2	U91836	11/12/2019 6:11:30 PM	N31739 N6072 N63045 N23979 N35656 N43353 N8129...	N20678-0 N39317-0 N58114-0 N20495-0 N42977-0 N...

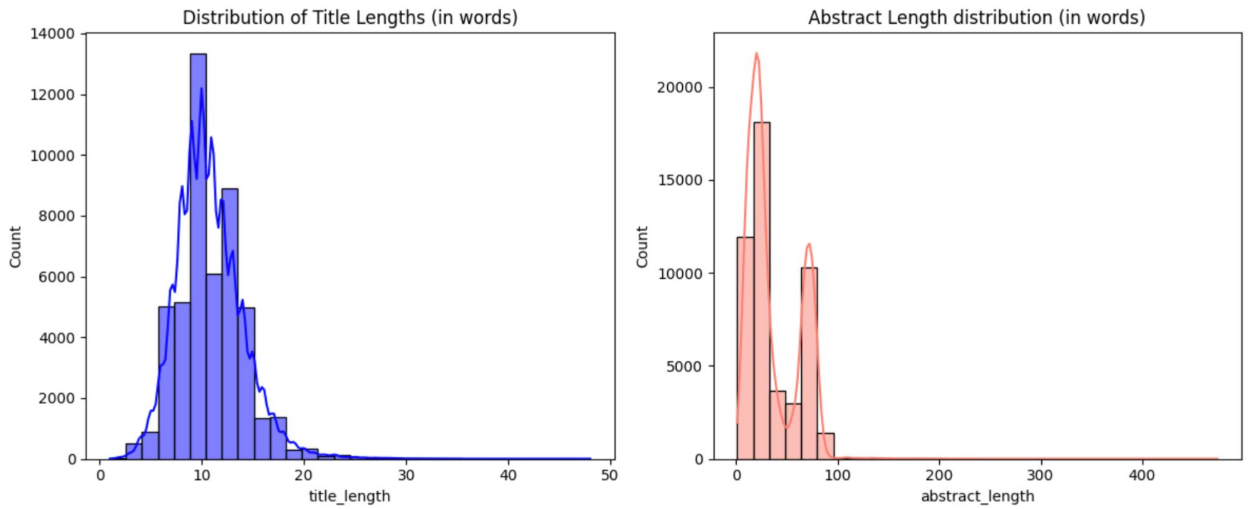
- News.tsv - The news.tsv contains the detailed information of news articles involved in the behaviors.tsv file. 51282 rows

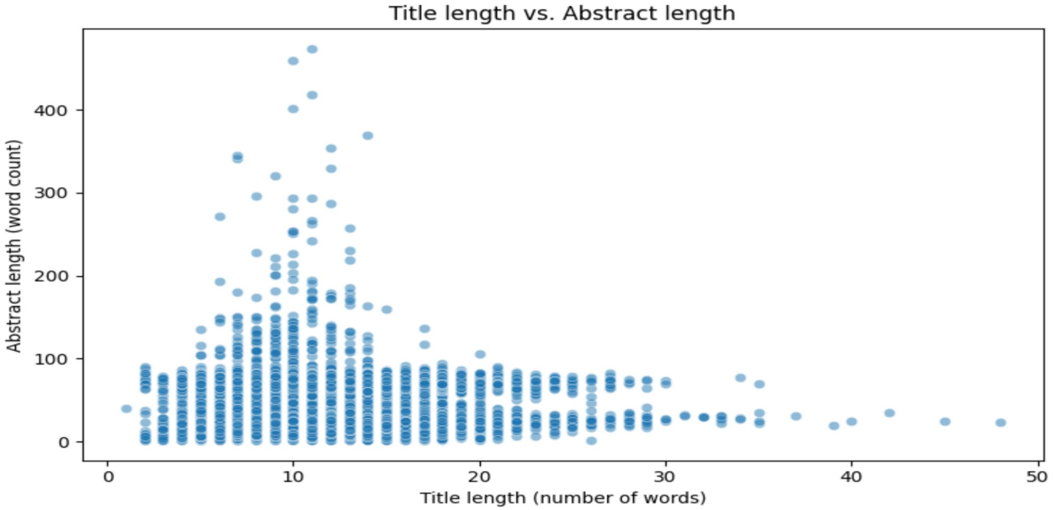
```
news.head(5)
```

	nid	category	subcategory	title	abstract	url	title_entities	abstract_entities
0	N55528	lifestyle	lifestyleroyals	The Brands Queen Elizabeth, Prince Charles, an...	Shop the notebooks, jackets, and more that the...	https://assets.msn.com/labs/mind/AAGH0ET.html	[{"Label": "Prince Philip, Duke of Edinburgh",...	[]
1	N19639	health	weightloss	50 Worst Habits For Belly Fat	These seemingly harmless habits are holding yo...	https://assets.msn.com/labs/mind/AAB19MK.html	[{"Label": "Adipose tissue", "Type": "C", "Wik...	[{"Label": "Adipose tissue", "Type": "C", "Wik...

Reference: https://www.kaggle.com/datasets/arashnic/mind-news-dataset/data

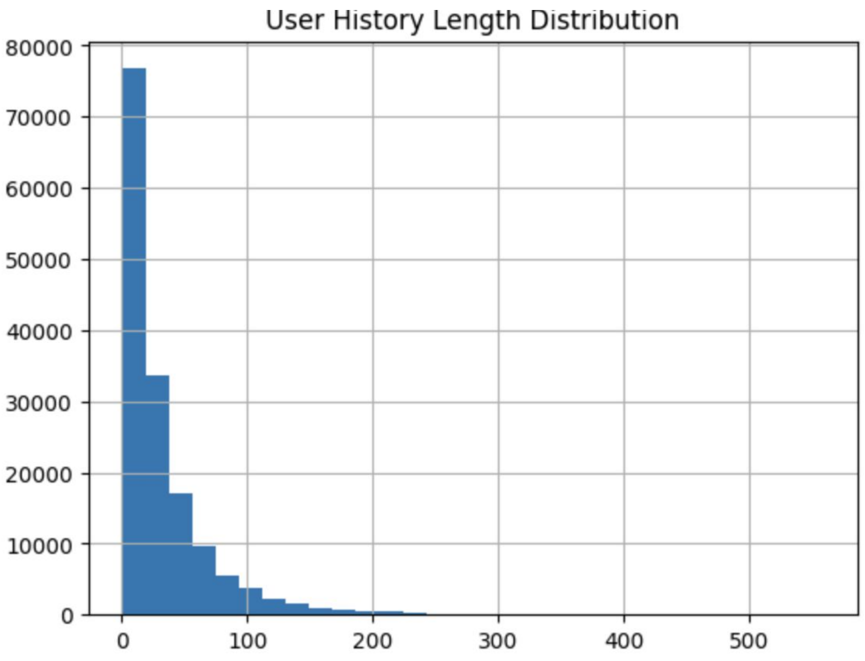
Data Analysis-News Articles

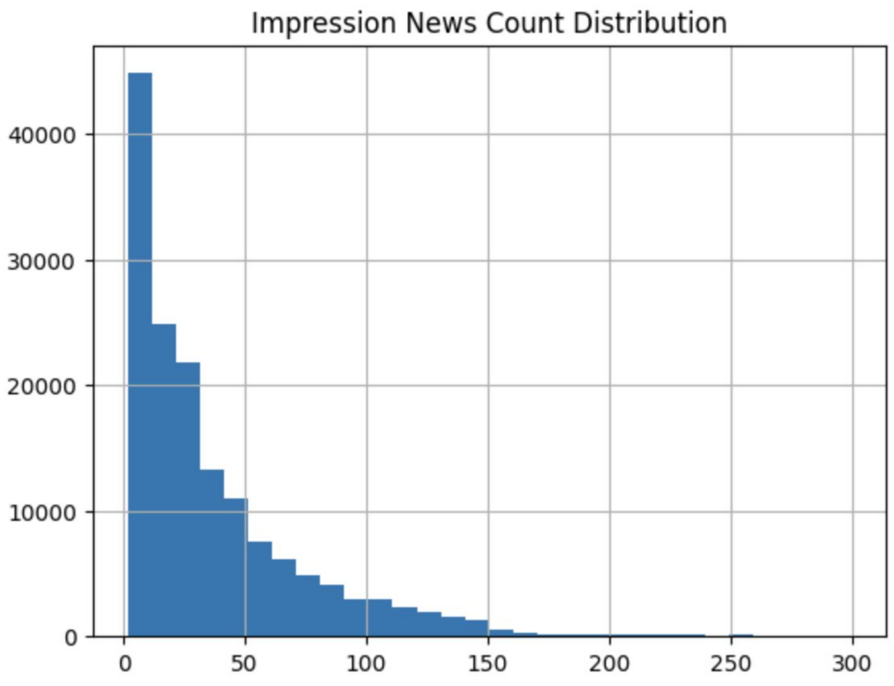




- Titles and abstracts are typically short.
- Authors will typically try to keep these corpii dense with information and features.

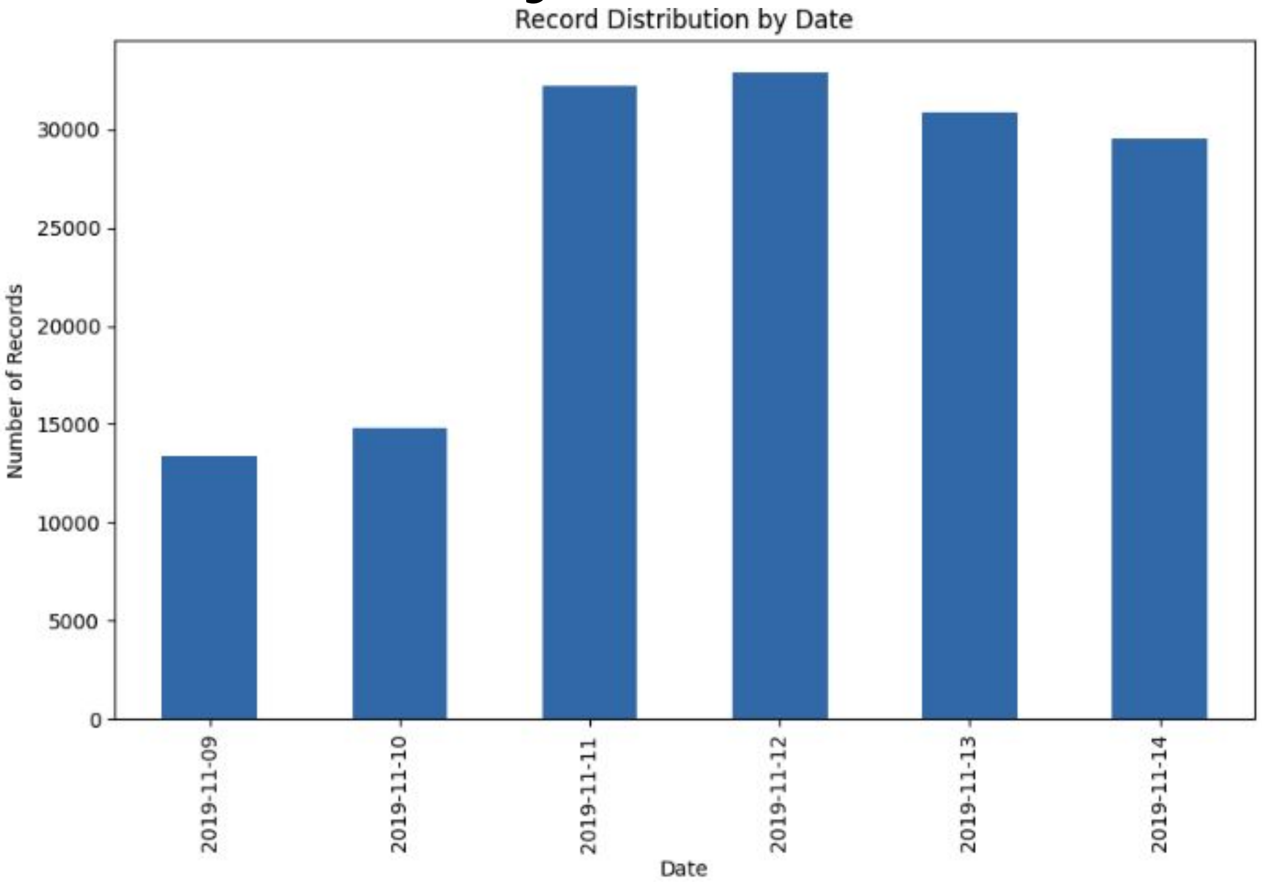
Data Analysis-Behaviors





- Although skewed towards 0 in both user and news interactions, overall interactions were not sparse.

Data Analysis-Behaviors



Record count by date:

2019-11-09	**13414**	
2019-11-10	**14815**	
2019-11-11	**32229**	context
2019-11-12	**32878**	
2019-11-13	**30893**	
2019-11-14	29498	evaluation

Data-Preprocessing

Null value filling

Replace the Title and Abstract with.fillna(" ")
Concatenation forms the full_text field for feature extraction

Standardized text format

To lowercase: text.lower()
Remove extra Spaces and line breaks: re.sub(r'\s+', ' ', text)
Remove punctuation marks (such as ' and -)

Participles and stop words

Use nltk.word_tokenize() for word segmentation
Remove the stop words in English (such as the, is, and, etc.)

Stem extraction (Stemming)

PorterStemmer: Reduce the dimension (such as running → run)

	News_ID	Category	SubCategory	Title	Abstract	full_text_clean
0	N55528	lifestyle	lifestyleroyals	The Brands Queen Elizabeth, Prince Charles, an...	Shop the notebooks, jackets, and more that the...	brand queen elizabeth princ charl princ philip...
1	N19639	health	weightloss	50 Worst Habits For Belly Fat	These seemingly harmless habits are holding yo...	50 worst habit belli fat seemingli harmless ha...
2	N61837	news	newsworld	The Cost of Trump's Aid Freeze in the Trenches...	Lt. Ivan Molchanets peeked over a parapet of s...	cost trump 's aid freez trench ukrain 's war l...
3	N53526	health	voices	I Was An NBA Wife. Here's How It Affected My M...	I felt like I was a fraud, and being an NBA wi...	nba wife 's affect mental health felt like fra...
4	N38324	health	medical	How to Get Rid of Skin Tags, According to a De...	They seem harmless, but there's a very good re...	get rid skin tag accord dermatologist seem har...

```
behaviors_clean
```

	Impression_ID	User_ID	Time	History	Impressions	Impressions_parsed	History_parsed	history_len	impr_num
0	1	U13740	11/11/2019 9:05:58 AM	N55189 N42782 N34694 N45794 N18445 N63302 N104...	N55689-1 N35729-0	[(N55689, 1), (N35729, 0)]	[N55189, N42782, N34694, N45794, N18445, N6330...	9	2
1	2	U91836	11/12/2019 6:11:30 PM	N31739 N6072 N63045 N23979 N35656 N43353 N8129...	N20678-0 N39317-0 N58114-0 N20495-0 N42977-0 N...	[(N20678, 0), (N39317, 0), (N58114, 0), (N2049...	[N31739, N6072, N63045, N23979, N35656, N43353...	82	11

Methodology

Methods Summary

- ➔ Pre-processing
- ➔ Baseline Models
  - ◆ Content-based
  - ◆ Collaborative Filtering
- ➔ Hybrid Models
  - ◆ TF-IDF+SBERT+NCF
  - ◆ SentiSBERT-KG-GNN
  - ◆ SentiSBERT-LSTM
  - ◆ SentiSBERT-GRU

Cold-Start Cases

- ◆ New User
- ◆ New Database

Methodology - Baseline (Content Base)

TF-IDF: Encodes text based on word importance using frequency.

Bert: Captures word context using deep transformers.

Sentence-Bert: Generates sentence embeddings for fast similarity matching.

Text features: `Title` + Abstract + Category

Tools: `sklearn.feature_extraction.text.TfidfVectorizer`
`transformers.BertModel`
`sentence_transformers.SentenceTransformer`

Parameter:
`max_features=7000`(Optimize to obtain the optimal value)
`BertModel.from_pretrained('bert-base-uncased')`
`SentenceTransformer('all-MiniLM-L6-v2')`

Recommendation method: Calculate the **cosine similarity** between the average vector of the user's historical news and the candidate news, and take the Top-K as the Top-10

TF-IDF	BERT	**SBERT**
- Precision@10: 0.0638	- Precision@10: 0.0635	**- Precision@10: 0.0696**
- Recall@10: 0.5009	- Recall@10: 0.4991	**- Recall@10: 0.5406**
- Mrr@10: 0.2463	- Mrr@10: 0.2311	**- Mrr@10: 0.2794**
- Ndcg@10: 0.2932	- Ndcg@10: 0.2824	**- Ndcg@10: 0.3277**
- Hit_rate@10: 0.5721	- Hit_rate@10: 0.5693	**- Hit_rate@10: 0.6130**

Methodology - Baseline (Content Base)

Can we use Hybrid embedding methods? TF-IDF+SBert

Hybrid content representations are effective for capturing both lexical and semantic characteristics of items.

TF-IDF captures **surface-level keyword matching**

SBERT provides **deep semantic understanding**

One-hot category adds **explicit structured information**

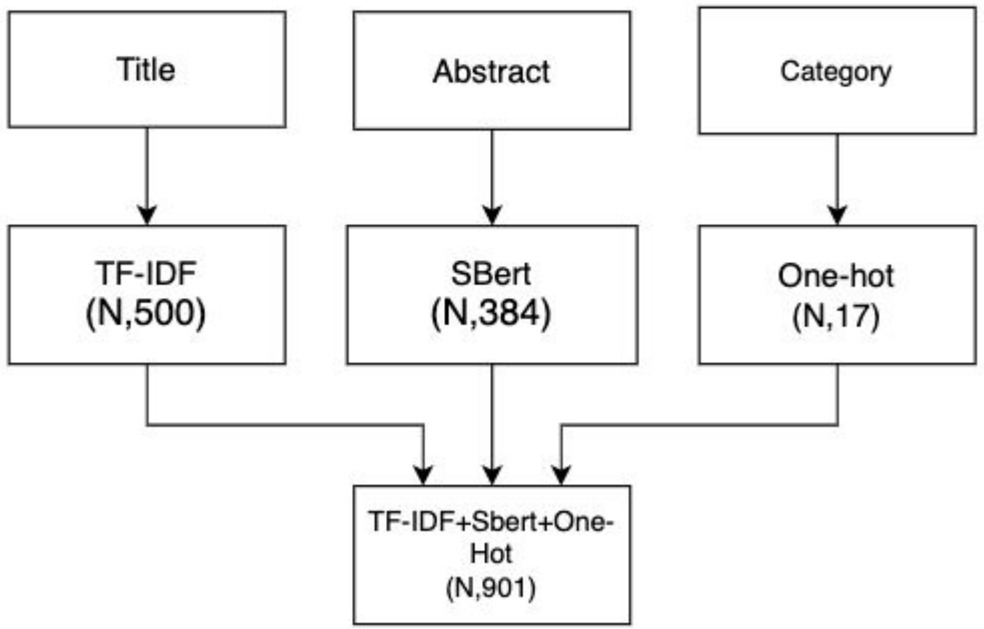


Figure:Hybrid Text Representation (TF-IDF + SBERT)

Parametes:

TfidfVectorizer(max_features=500) to embedding Title

SentenceTransformer('all-MiniLM-L6-v2') to embedding Abstract

One-hot to embedding Category

TF-IDF+SBERT
Precision@10: 0.0692
Recall@10: 0.5481
MRR@10: 0.2845
NDCG@10: 0.3351
Hit@10: 0.6152

Methodology - Baseline(Collaborative Filtering)

- Neural Collaborative Filtering (NCF) was introduced by He et al. (2017) to capture **non-linear interactions** between users and items through **multi-layer perceptrons (MLPs)**.

- Unlike traditional matrix factorization which models dot-product interactions, NCF allows the model to learn more complex patterns from data.

- The model uses **embedding layers** to represent users and items in a shared latent space, and **deep networks** to predict user preference scores.

Parameters:

epoch=10

optimizer = torch.optim.Adam(model.parameters(), lr=0.001)

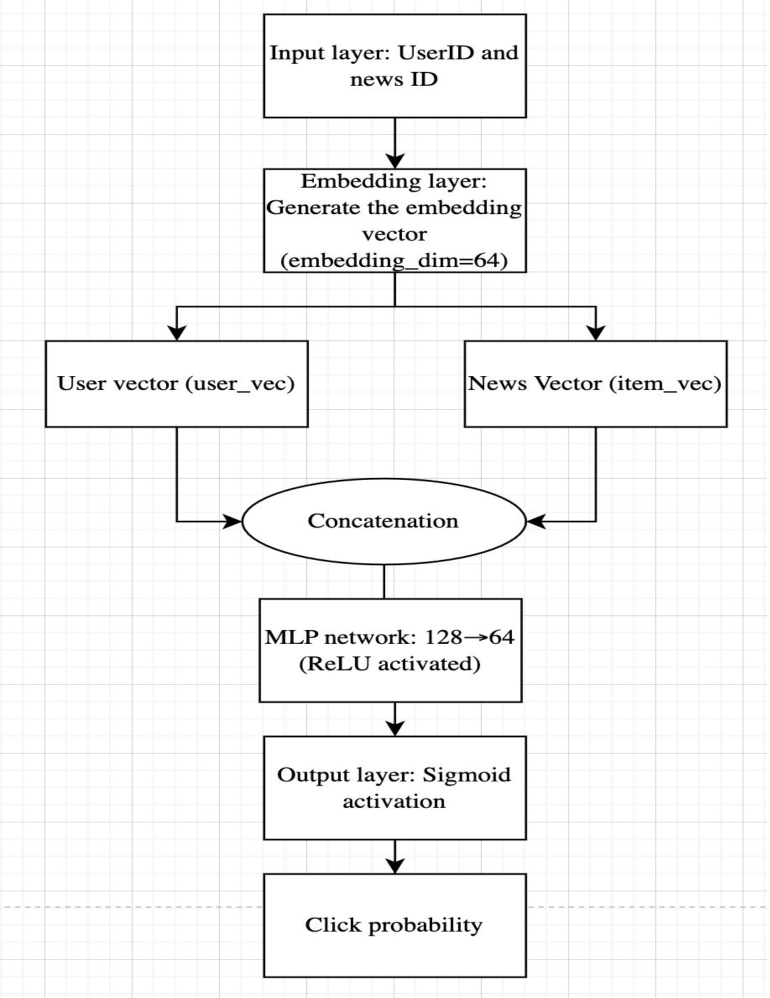
criterion = nn.BCELoss()

EarlyStopping

best_val_loss = float('inf')

patience = 3

StandardScaler()



He, X., Liao, L., Zhang, H., Nie, L., Hu, X., & Chua, T.-S. (2017). ***Neural Collaborative Filtering****. In Proceedings of the 26th International Conference on World Wide Web (WWW '17), pp. 173–182.*

Methodology - Hybrid model 1: TF-IDF+SBERT+Neural CF

Hybrid models have been shown to effectively combine semantic understanding from content with behavioral signals from interactions (Burke, 2002).

Content-based representations from TF-IDF and SBERT, and User embeddings learned via Neural Collaborative Filtering.

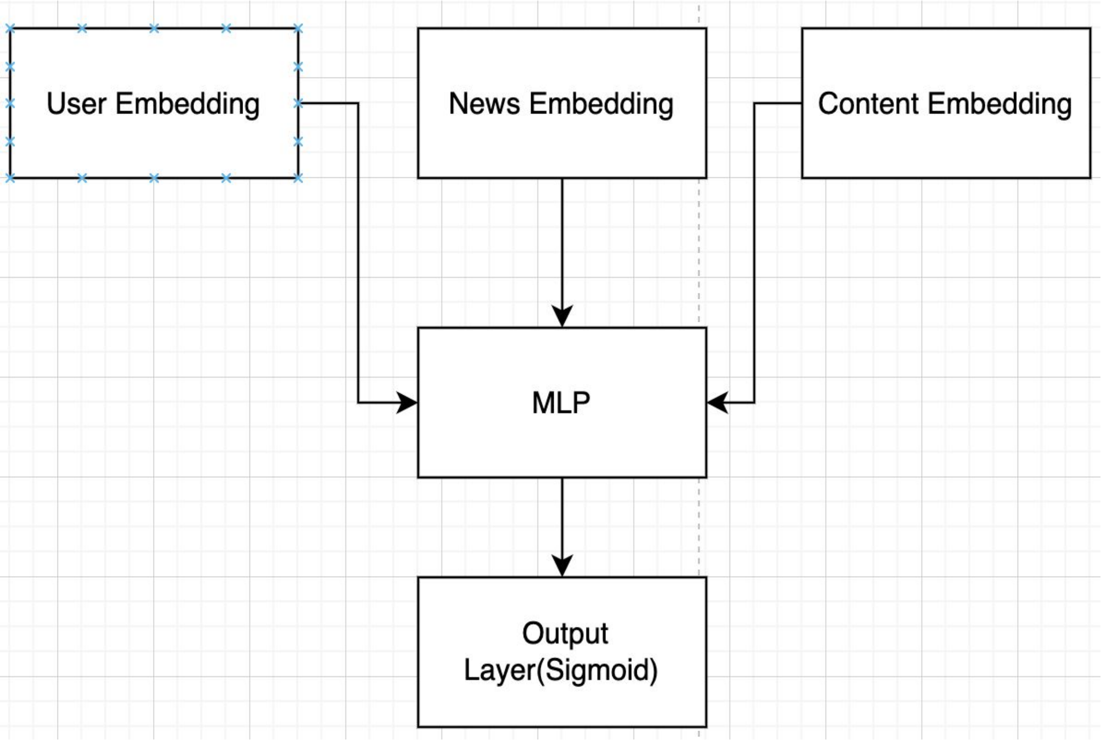


Figure:Hybrid Recommendation Network

Parameters:
TF-IDF: max_features=500(best parameters)
SBERT: all-MiniLM-L6-v2
Standardization tool: StandardScaler()
Embedding layer dimension: Embedding(dim=32)
Content feature layer (CB): Linear(901 → 64)
Fully connected layer: 128 → 64 → 1
Optimizer: Adam(lr=0.001)
Loss function: BCELoss()
Batch Size: 16
Epochs: 10

Burke R. Hybrid recommender systems: Survey and experiments[J]. User modeling and user-adapted interaction, 2002, 12(4): 331-370.

Drawbacks of above established methods

- Baseline: Content base:

  Over-specialisation

  User cold-start unsolved

  Feature sparsity & noise

  No community signal

  lack of diversity

- Baseline: Collaborative filtering

  User & item cold-start

  Data sparsity

  Popularity bias

  lack of diversity

- Hybrid model1:TD-IDF + SBERT + NCF

  Feature-level incompatibility

  Explainability drops

  lack of diversity

  But has strengths

Literature Review- Current studies

- **RippleNet (Wang et al.,CIKM ’18)**

  KG preference propagation tackles sparsity, yet relies on pre-defined hop depth.

- **KGAT (Wang et al.,KDD ’19)**
  Attention over high-order KG paths; ignores rich text semantics.

- **HGNN-News (Hu et al., WWW ’20)**

  Heterogeneous GNN for user–news–entity graphs, but no sentiment signal.

- **UD GNN (Zhu et al.,ICDE ’21)**
  Disentangles neighbours for news rec; still pure click-based.

No model fuses *sentence-level semantics (SBERT)*, *light-weight sentiment*, and *KG attention* in a single end-to-end framework.
Our answer：SentiSBERT-KG-GNN integrates all three, enabling cold-start coverage, interpretability and richer diversity.

Hybrid Model 2:

Our State-of-Art Model Invention:

SentiSBERT-Knowledge Graph-Graph Neural Network Model

(SentiSBERT-KG-GNN)

Strengths of SentiSBERT-KG-GNN

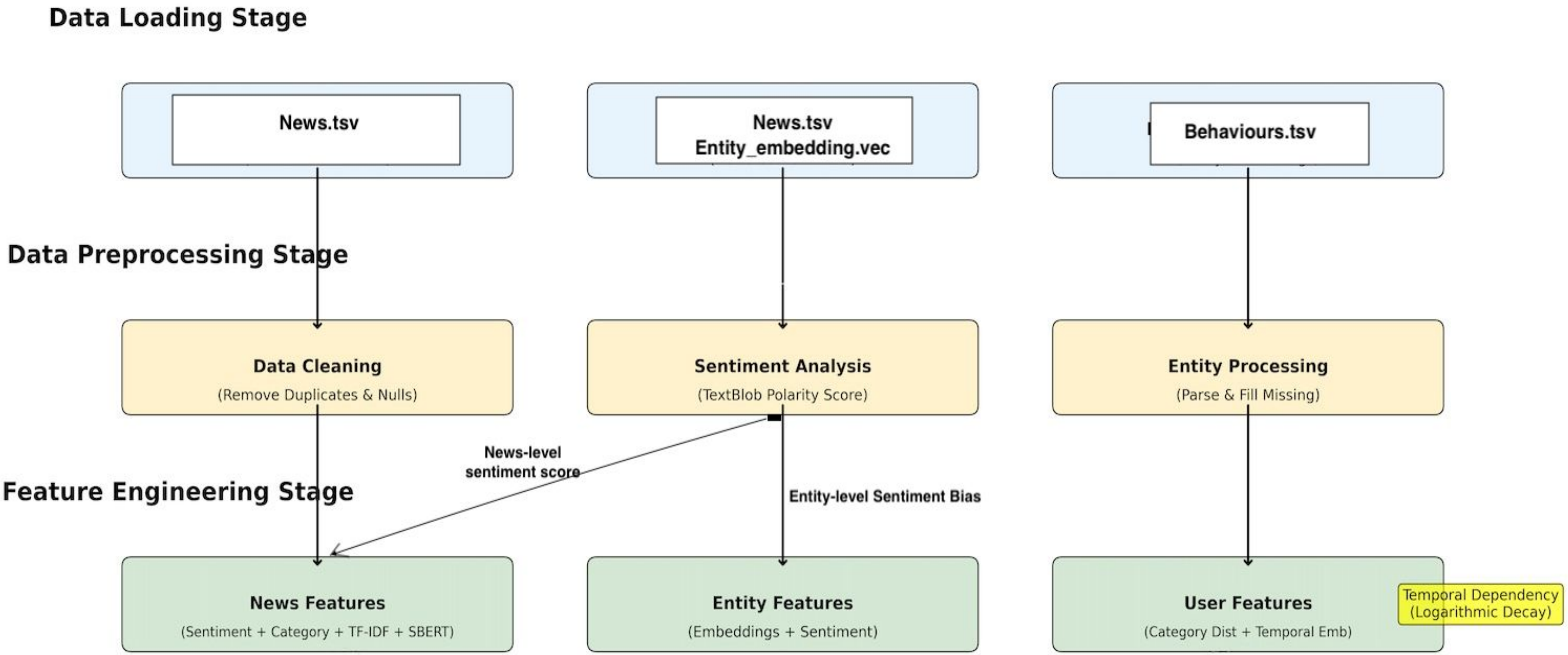
It tackles:

1. **Boosted rich features than flat interaction graphs**
2. **Natural cure for sparsity and cold-start**
3. **High-order semantics with limited parameters**
4. **Built-in interpretability**
5. **Multi-Modular feature fusion:** It can plug various text vectors, sentiment scalars and temporal dependency straight into the same GNN
6. **Increase diversity**
7. **Build up a personalized recommendation system**

Together, KG-GNN gives us expressiveness, cold-start robustness, transparency, and parameter efficiency in one coherent framework.

Implementation

Feature Engineering



Feature Engineering

Sentiment Embedding

Sentiment-Aware Knowledge Graph: We construct a heterogeneous graph connecting users, news articles, and entities, where entity nodes incorporate sentiment information derived from news titles. This enables the model to understand not just what entities are mentioned, but how they are portrayed emotionally.

Feature Engineering

Semantic Embeddings

Multi-Modal Feature Fusion: We combine SBERT semantic embeddings (384-dim), TF-IDF text features (100-dim), sentiment scores, and categorical features using one-hot to create rich 502-dimensional news representations that capture both semantic and affective aspects of content.

Feature Engineering

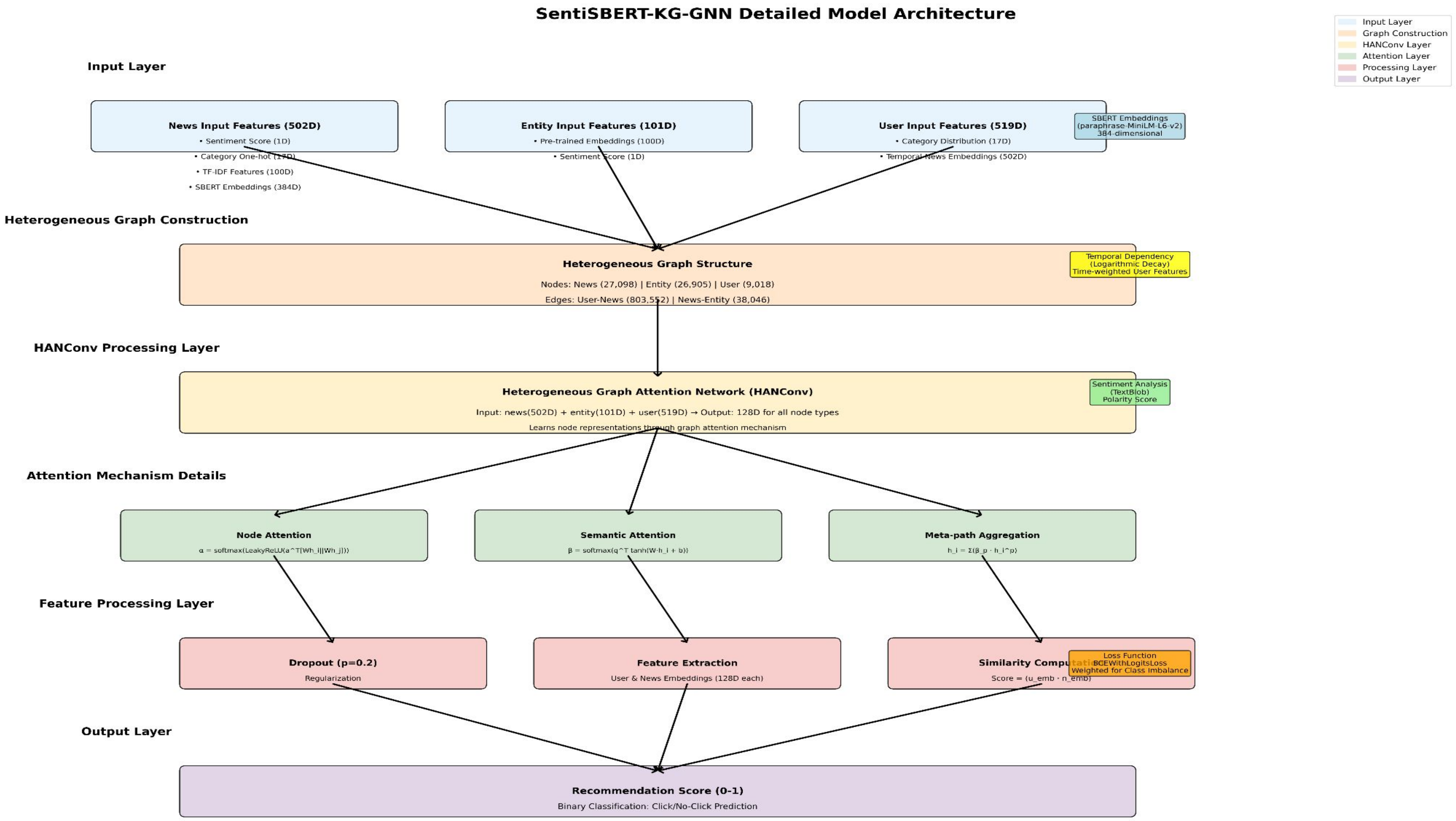
Temporal Dependency

Temporal Dependency Modeling: We implement logarithmic temporal decay to weight user behaviors, ensuring recent interests have higher influence while preserving historical preferences. This addresses the dynamic nature of user interests over time.

Logarithmic temporal decay log-based：

$$w_t = \frac{1}{\log(c + \Delta t)}$$

Here, Δt represents the time difference between the current time and the user's behavior, and c is a constant set to avoid the denominator being zero



SentiSBERT-KG-GNN Detailed Model Architecture
Input Layer
Graph Construction
HANConv Layer
Attention Layer
Processing Layer
Output Layer
Input Layer
News Input Features (502D)
• Sentiment Score (1D)
• Category One-hot (17D)
• TF-IDF Features (100D)
• SBERT Embeddings (384D)
Entity Input Features (101D)
• Pre-trained Embeddings (100D)
• Sentiment Score (1D)
User Input Features (519D)
• Category Distribution (17D)
• Temporal News Embeddings (502D)
SBERT Embeddings (paraphrase-MiniLM-L6-v2) 384-dimensional
Heterogeneous Graph Construction
Heterogeneous Graph Structure
Nodes: News (27,098) | Entity (26,905) | User (9,018)
Edges: User-News (803,552) | News-Entity (38,046)
Temporal Dependency (Logarithmic Decay) Time-weighted User Features
HANConv Processing Layer
Heterogeneous Graph Attention Network (HANConv)
Input: news(502D) + entity(101D) + user(519D) → Output: 128D for all node types
Learns node representations through graph attention mechanism
Sentiment Analysis (TextBlob) Polarity Score
Attention Mechanism Details
Node Attention
α = softmax(LeakyReLU(a^T[Wh_i||Wh_j]))
Semantic Attention
β = softmax(q^T tanh(W·h_i + b))
Meta-path Aggregation
h_i = Σ(β_p · h_i^p)
Feature Processing Layer
Dropout (p=0.2)
Regularization
Feature Extraction
User & News Embeddings (128D each)
Similarity Computation
Score = (u_emb · n_emb)
Loss Function BCEWithLogitsLoss Weighted for Class Imbalance
Output Layer
Recommendation Score (0-1)
Binary Classification: Click/No-Click Prediction

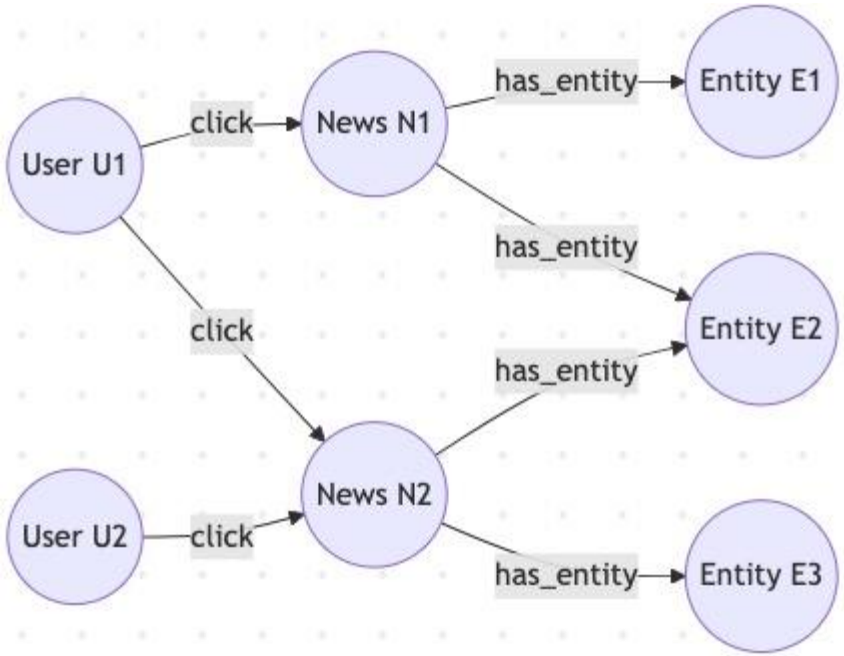
Heterogeneous Graph Attention Network

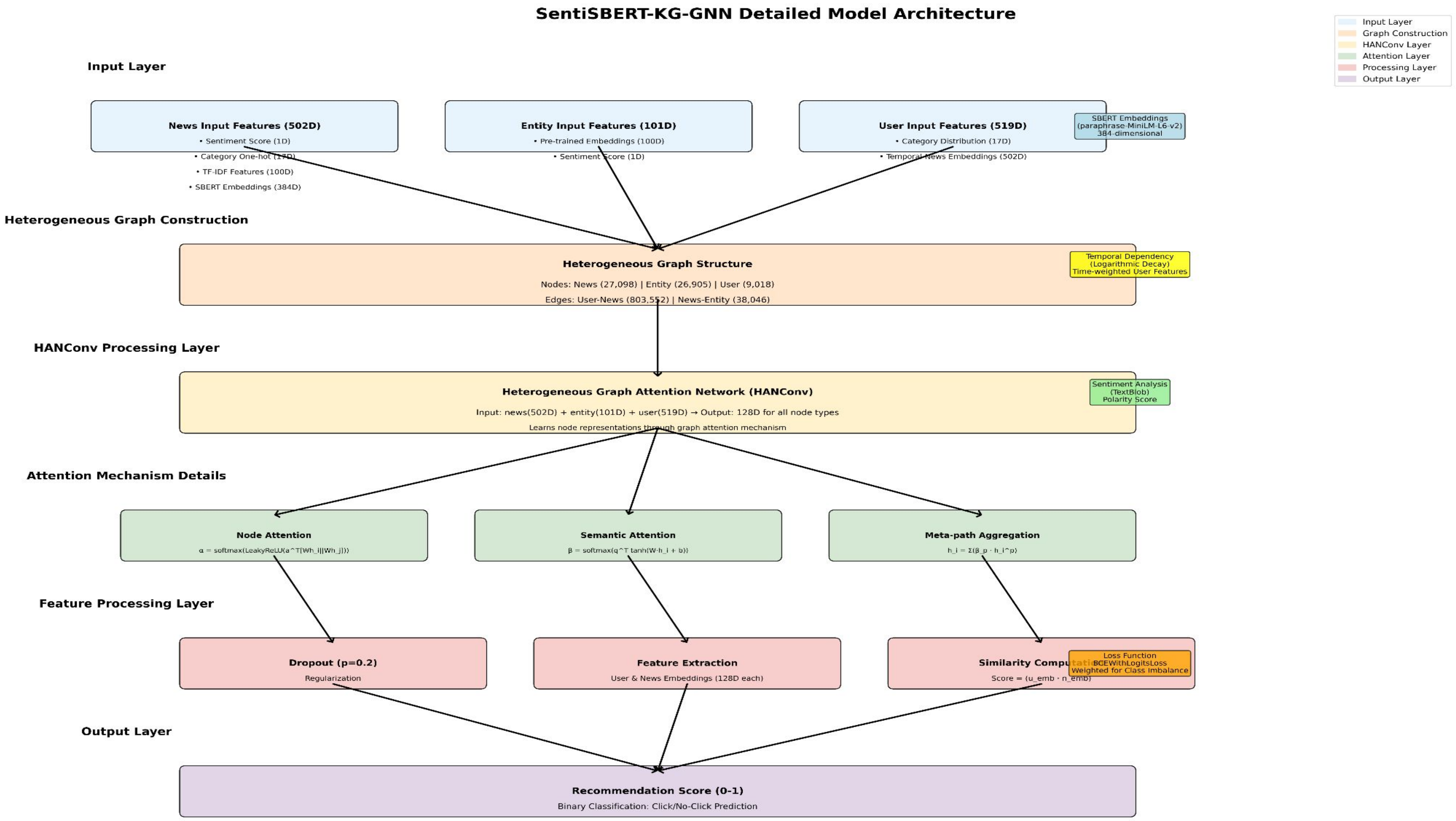
Graph Structure and Information Flow:

- User-News Edges (clicked)
- News-Entity Edges (has_entity)

HANConv layer:

- user → news attention: Learns which users' behaviors are most relevant for understanding news appeal
- news → entity attention: Learns which entities are most important for characterizing news content
- entity → news attention: Propagates entity information back to news nodes





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Loss Function BCEWithLogitsLoss Weighted for Class Imbalance
Output Layer
Recommendation Score (0-1)
Binary Classification: Click/No-Click Prediction

Hyperparameters & Tuning

Parameters set up：

- 10 epoch
- Postive samples/negative samples deployed during training：1:4

Hyperparameter tuning using GridSearch on subset-samples：

Feature Engineering:

- TF-IDF+SBERT parameters: StandardScaler normalization
- temporal_decay_factor: The best param is 12
- temporal_function = 'logarithmic'

Hancock model:

- dropout_rate: best param ：0.2
- learning_rate: best param 0.001
- Hancov model‘s Hiddem dim：256

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Key Innovation

- Sentiment-Aware: Sentiment analysis enhanced news and entity representations
- Multi-Modal Fusion: Fusing TF-IDF, SBERT, one-hot category, and sentiment features
- Temporal Dependency: Logarithmic time decay weighting for user behaviors
- Heterogeneous Graph: User-News-Entity triplet knowledge graph structure
- GNN Attention Mechanism: Multi-head attention learning importance across nodes

Other techniques：

Derived from the inspiration of our SentiSBERT-KG-GNN model

- Hybird model3： SentiSBERT-GRU
- Hybrid model4:   SentiSBERT-LSTM

Hybrid Model 3&4

Hybrid with Sequential Models:

- Hybrid Model 3：SentiSBERT-GRU: fewer params, faster to converge
- Hybrid Model 4：SentiSBERT-LSTM: captures longer context, slower inference

Order Awareness:

- NCF models static user–item affinities only.
- GRU / LSTM capture the temporal order of clicks → aligns with users' current intent.

Context Dependency:

- GRU / LSTM learn semantic relationships & time gaps between news items.
- NCF lacks session-level context modelling.

Hybrid Model 3&4

Trade-off

- GRU is lighter & faster
- LSTM retains longer context.

When to use

- Real-time / resource-limited → choose **GRU**
- Rich long-term patterns → choose **LSTM**

Metric highlights

- Precision@10 +6 % (LSTM over GRU)
- Inference latency –24 % (GRU over LSTM)

Cold Start - New User

Top Popular News

- Suggest top-N popular news articles

Top Popular Entities

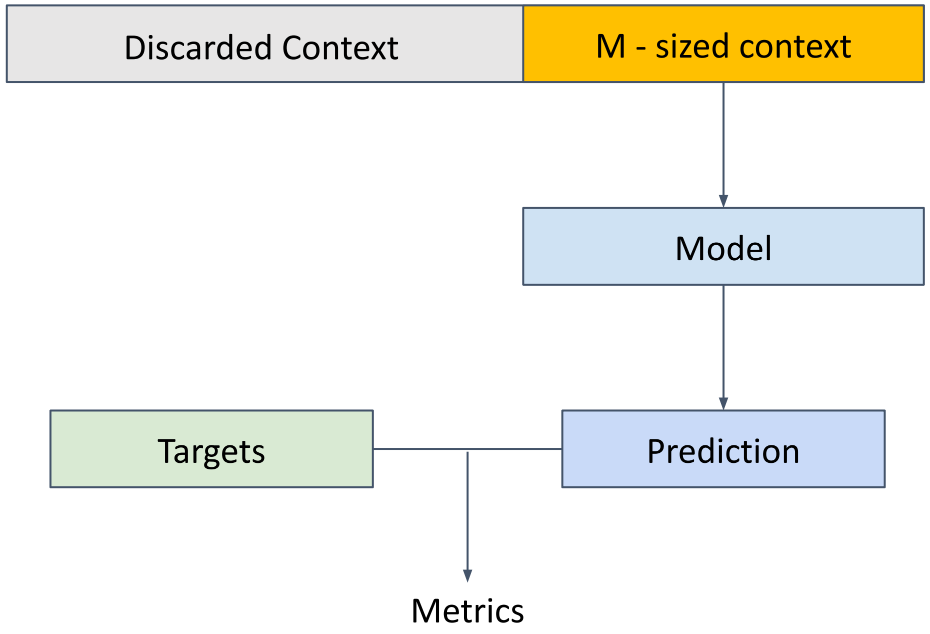
- Suggest the most popular topics
- Pipeline:
  - Collect most popular topics from other users
  - Ranking news articles by maximum overlap with most popular topics

	rank	news_id	title	frequency
0	1	N42620	Heidi Klum's 2019 Halloween Costume Transformation Is Mind-Blowing But, Like, What Is It?	1372
1	2	N306	Kevin Spacey Won't Be Charged in Sexual Assault Case After Accuser Dies	1362
2	3	N35671	Donald Trump Jr. reflects on explosive 'View' chat: 'I don't think they like me much anymore'	1143
3	4	N4607	Cause determined in Jessi Combs' fatal speed record crash	1123
4	5	N18870	Here Are the Biggest Deals We're Anticipating for Black Friday	1081
5	6	N28257	Meghan McCain confronts Trump Jr.: 'You and your family have hurt a lot of people'	1068
6	7	N64273	Watch: 'The View' Versus Donald Trump Jr.: Loud, Low Blows, Politics, Scandals And Great TV	989
7	8	N22816	Snow crab sells for record price in Japan	983
8	9	N8148	Holocaust survivor under guard amid death threats	882
9	10	N47020	The News In Cartoons	858

Cold Start - New Database

New Database Simulation

- To test our models against a cold-start situation, an empty dataset was simulated.
- For each user, their "history" or context set is limited to their most recent M items. This term is defined as user-history-max-size in charts.
- For small Ms, simulates situation where there has not been adequate time for the database to mature.



Evaluation - Metrics

Priority Metrics

- **nDCG @10** is our lead metric. It is the de-facto headline number in almost every MIND-based study (e.g. NAML, NRMS, KGAT-News, Hyper4NR) because it rewards *both* relevance and correct ranking within the first screen of results.

- **Hitrate / Recall @10** is the second priority. Industry teams (MS-News, Google News) track it to make sure at least one click-worthy article appears; many papers report it alongside nDCG.

Other Metrics

- Precision@10
- MRR@10

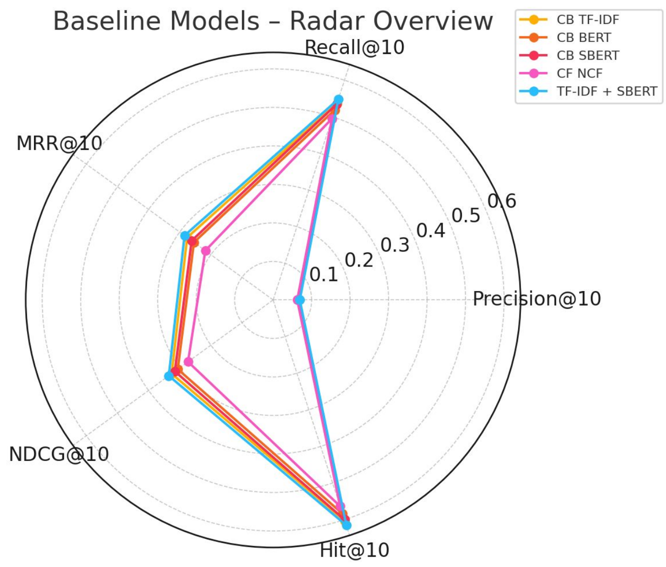
Results and Discussion

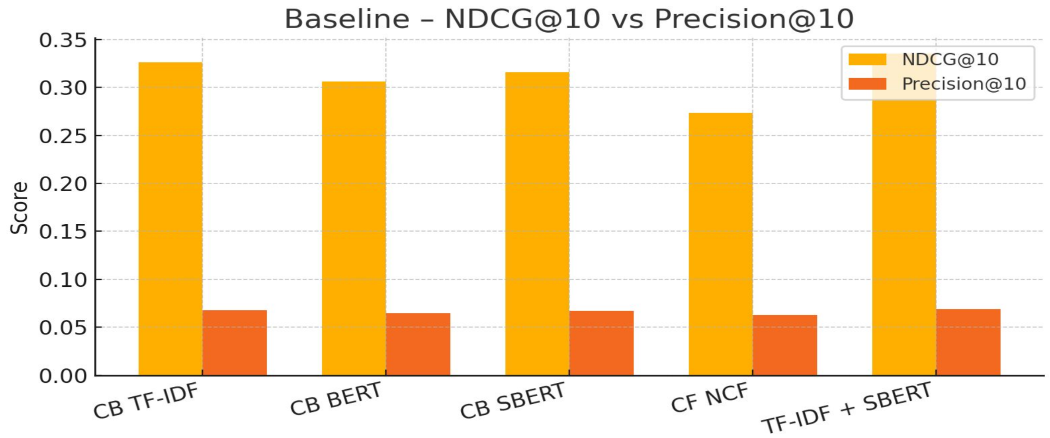
- Baseline Models
- Hybrid Models
- Cold Start Analysis

Results - Baselines

- Five baselines evaluated: CB TF-IDF/BERT/SBERT, CF NCF, TF-IDF + SBERT
- **TF-IDF + SBERT** is the best baseline on NDCG & Precision、recall
- **CF NCF underperforms** content models on NDCG & Precision、recall
- Next steps:Integrate baseline model with collaborative filtering, knowledge graph & multimodal features to raise ranking quality

Model	Precision@10	Recall@10	MRR@10	NDCG@10	Hit@10
CB TF-IDF	0.0676	0.5369	0.2761	0.3259	0.6038
CB BERT	0.0648	0.5190	0.2543	0.3062	0.5840
CB SBERT	0.0669	0.5344	0.2617	0.3156	0.5997
CF NCF	0.0626	0.4952	0.2178	0.2733	0.5632
TF-IDF + SBERT	0.0692	0.5481	0.2845	0.3351	0.6152

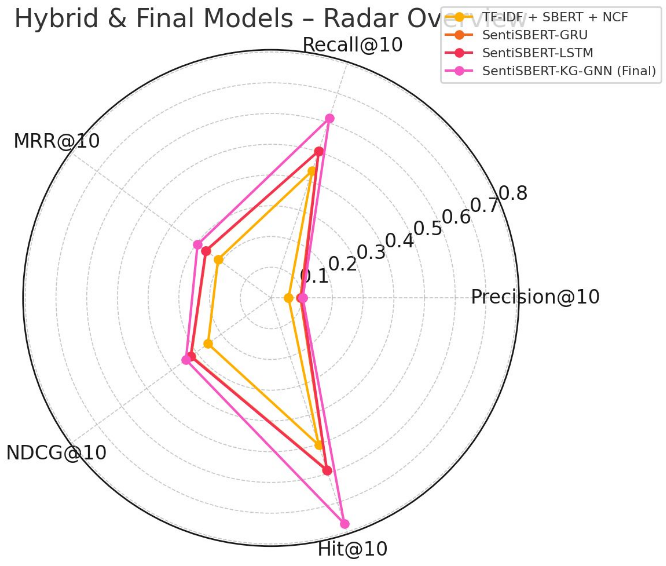


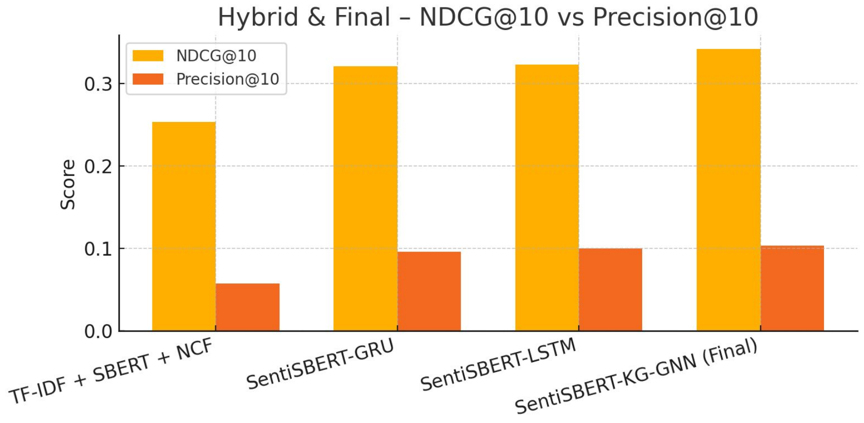


Results - Hybrid

- SentiSBERT-KG-GNN (Final) achieves the highest scores across all metrics.
- Precision@10: 0.1034, ~3.5% relative improvement over SentiSBERT-LSTM (0.0999).
- Recall@10: 0.6143 (+22.0% vs. LSTM); Hit@10: 0.7716 (+30.6% vs. LSTM).
- Incorporating a knowledge graph and GNN yields substantial gains in both recommendation accuracy and coverage.

Model	Precision@10	Recall@10	MRR@10	NDCG@10	Hit@10
TF-IDF + SBERT + NCF	0.0571	0.4356	0.2132	0.2535	0.5031
SentiSBERT-GRU	0.0958	0.5026	0.2609	0.3213	0.5874
SentiSBERT-LSTM	0.0999	0.5031	0.2625	0.3235	0.5904
SentiSBERT-KG-GNN (Final)	0.1034	0.6143	0.2956	0.3423	0.7716





Results - Cold Start - New User

Top-popular comparison for new users (no data)

- Top-popular by news was the best solution for cold-starting a new user.

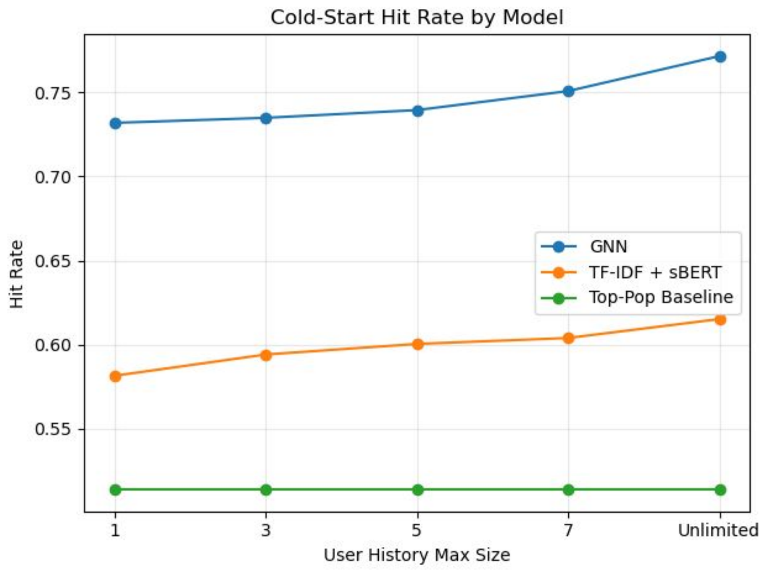
- Top-popular by entities did not perform well.
  - It was found that entities only captured *nouns* where as user interest in a “topic” may not be linked to that.
  - For example, a user interested in “crime” is something not captured by entities.

	hit_rate@N	precision@N	recall@N	MRR@N	NDCG@N
Top-pop News	**0.5138**	**0.06681**	**0.30386**	**0.203274**	**0.191022**
Top-pop Entity	0.419	0.05955	0.181658	0.139868	0.120097

Results - Cold Start - New Database

Results - New Database - Proposed Method

- Our proposed model sentiSBERT-KG-GNN performed well in a cold-start situation.
    - When user data is sparse, the attention mechanism was able to instead rely on more content-based features.



Hit Rate @ N when database size is limited			
User History Max Size	GNN	TF-IDF + sBERT	Top-pop baseline
1	**0.7319**	0.5814723503	0.5138
3	**0.7349**	0.5941376596	0.5138
5	**0.7395**	0.6004215265	0.5138
7	**0.7508**	0.6039537622	0.5138
Unlimited	**0.7716**	0.6152	0.5138

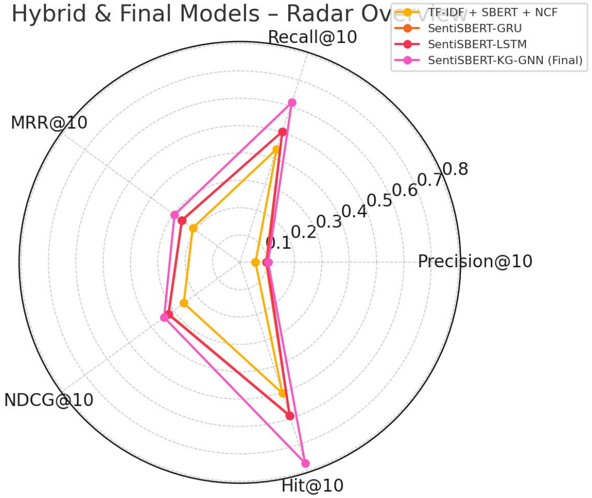
Conclusions

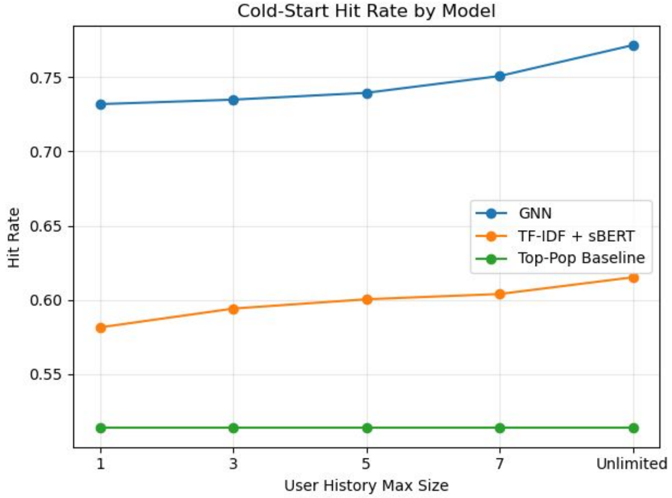
Conclusions

- **Identified** Modelling gap between cold-start performance of content-based and general performance of user-based methods in news recommendation

- **Combined** content-based and user-based data using our model **SentiSBERT-KG-GNN** to best leverage benefits of both approaches.

- **Achieved** design targets; having superior performance from a mix of user and content-based methods while retaining great cold-start performance..





Key Charts demonstrating (1) superior performance to content-based methods and (2) retaining their immunity to cold-starts.

Future work

- Relation Embeddings for Edge Features
- Dual-Field Sentiment: Extend sentiment embeddings to both title and abstract entities
- Refine Model Architecture. Explore hierarchical fusion (content + behavior + KG), and dynamic weight mechanisms
- Incorporate Transformer-based Sequence Models


Questions